



Introduction to Malware

? What is Malware?

- Malware = Malicious Software
- Designed to damage, steal, spy, or control systems
- Can infect computers, mobiles, servers, and networks
- Often runs without user knowledge

Why Hackers Use Malware

- Steal personal or financial data
- Gain remote access to systems
- Spy on users (keylogging, webcam access)
- Spread attacks to other systems
- Demand money (ransomware)

How Hackers Perform Malware Attacks

- Phishing emails with malicious attachments
- Fake or cracked software downloads
- Infected USB drives
- Malicious websites and ads
- Exploiting unpatched system vulnerabilities



Types of Malware

Virus

- Attaches to legitimate files
- Spreads when infected file is executed
- Requires user interaction

Worm

- Self-replicates automatically
- Spreads through networks
- No user interaction required

RAT (Remote Access Trojan)

- Allows attacker full remote control
- Can access files, camera, keyboard
- Often used for spying

Rootkit

- Hides deep inside the system
- Avoids detection by antivirus
- Maintains long-term unauthorized access

Trojan

- Disguised as legitimate software
- Creates backdoors for attackers
- Does not self-replicate



How to Prevent Malware

- Install software only from trusted sources
- Avoid clicking unknown links or attachments
- Keep OS and software updated
- Use strong passwords
- Enable firewall and antivirus
- Do not use cracked or pirated software

Malware Detection Tools

- Windows Defender
- Malwarebytes
- ClamAV (Linux)
- Antivirus with real-time protection



Malware Analysis Tools

Static Analysis (Without Running File)

- VirusTotal, Strings, PEStudio & Detect It Easy (DIE)

Dynamic Analysis (Run in Safe Environment)

- Any.Run, Hybrid Analysis, Cuckoo Sandbox, Process Monitor & Wireshark

Important Points

- Malware ≠ hacking tools
- Most attacks succeed due to user mistakes
- Misconfiguration and outdated systems increase risk
- Security is about prevention + detection + response